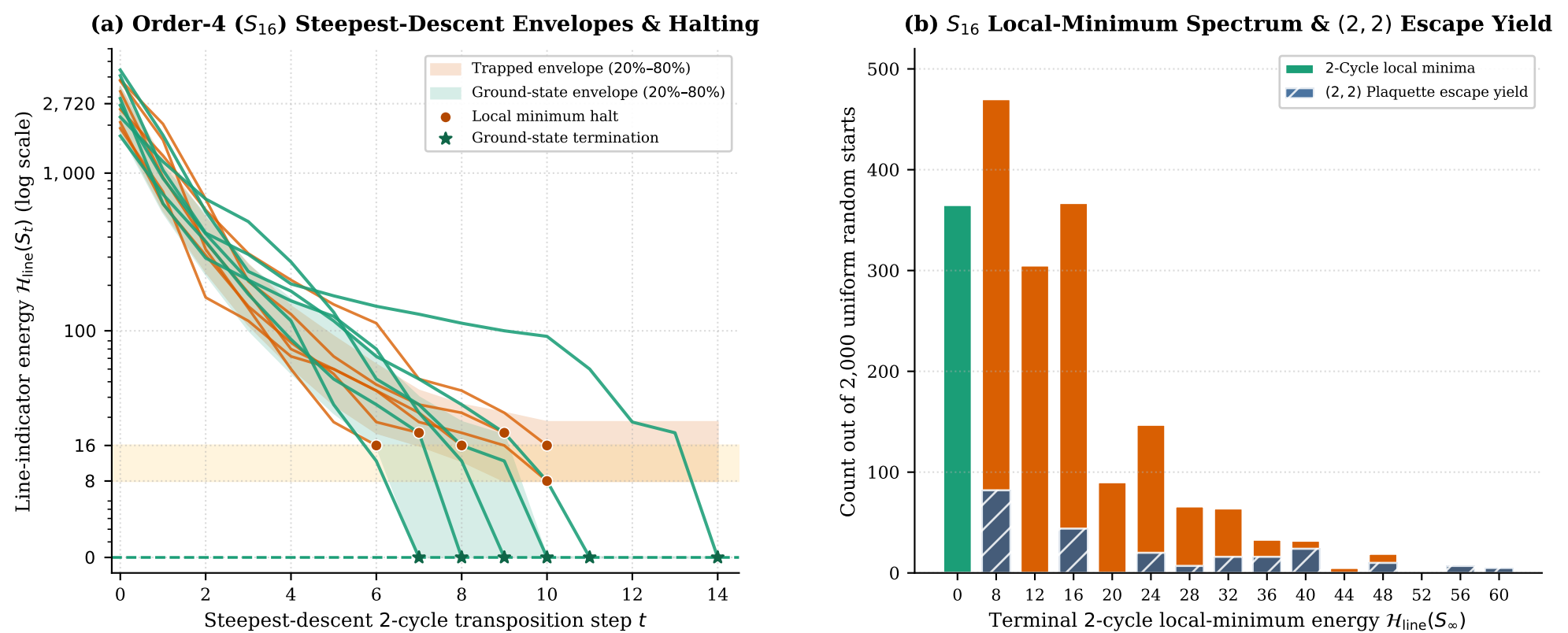
(a) Order-4 (S_{16}) Steepest-Descent Envelopes & Halting
Trapped envelope (20%–80%)
Ground-state envelope (20%–80%)
Local minimum halt
Ground-state termination
Line-indicator energy $\mathcal{H}_{\text{line}}(S_t)$ (log scale)
2,720
1,000
100
16
8
0
0
2
4
6
8
10
12
14
Steepest-descent 2-cycle transposition step t
(b) S_{16} Local-Minimum Spectrum & (2, 2) Escape Yield
2-Cycle local minima
(2, 2) Plaquette escape yield
Count out of 2,000 uniform random starts
500
400
300
200
100
0
0
8
12
16
20
24
28
32
36
40
44
48
52
56
60
Terminal 2-cycle local-minimum energy $\mathcal{H}_{\text{line}}(S_\infty)$